□ 1. Overall Docker Strategy

You will provide:

□ A **multi-container system** using Docker Compose with:

- MQTT Broker (Mosquitto)
- Node-RED (dashboard + flows)
- Python Edge AI environment
- (Optional) InfluxDB + Grafana

Students only need to run:

docker-compose up

□ 2. Project Structure

```
edge-ai-stack/
│
├── docker-compose.yml
├── python/
│   ├── Dockerfile
│   ├── mqtt_publisher.py
│   ├── edge_ai.py
│
├── mosquitto/
│   └── mosquitto.conf
│
└── node-red/
    └── flows.json
```

□ 3. docker-compose.yml

```
version: '3.8'

services:

  mqtt:
```

```
  image: eclipse-mosquitto:2.0
  container_name: mqtt
  ports:
    - "1883:1883"
  volumes:
    - ./mosquitto/mosquitto.conf:/mosquitto/config/mosquitto.conf

 node-red:
  image: nodered/node-red:latest
  container_name: node-red
  ports:
    - "1880:1880"
  volumes:
    - ./node-red:/data
  depends_on:
    - mqtt

 python-edge:
  build: ./python
  container_name: python-edge
  depends_on:
    - mqtt
```

□ 4. Python Dockerfile

File: `python/Dockerfile`

```
FROM python:3.11-slim

WORKDIR /app

COPY . /app

RUN pip install --no-cache-dir paho-mqtt numpy

CMD ["python", "mqtt_publisher.py"]
```

□ Students can switch to:

```
python edge_ai.py
```

□ 5. Mosquitto Config

File: `mosquitto/mosquitto.conf`

```
listener 1883
allow_anonymous true
```

(Simple for student use — no authentication)

□ 6. How to Run

Students run:

```
docker-compose up --build
```

Then open:

- Node-RED → http://localhost:1880
- MQTT → localhost:1883

□ 7. Running Multiple Python Scripts

Students can modify docker-compose to run AI instead:

```
python-edge:
 build: ./python
 command: python edge_ai.py
```

OR run manually:

```
docker exec -it python-edge python edge_ai.py
```

□ 8. AI-Ready Image (Better Version)

You can provide a more advanced image with AI libraries:

```
FROM python:3.11-slim

WORKDIR /app
COPY . /app

RUN pip install --no-cache-dir \
  paho-mqtt \
  numpy \
  pandas \
  scikit-learn \
  matplotlib

CMD ["python", "mqtt_publisher.py"]
```

 This allows:

- basic ML models
- anomaly detection
- data analysis

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9. Add-On: Grafana + InfluxDB

Add to `docker-compose.yml`:

```
influxdb:
 image: influxdb:2.7
 ports:
   - "8086:8086"

grafana:
 image: grafana/grafana:latest
 ports:
   - "3000:3000"
```

Now students get:

- Real-time dashboards
- Historical data storage

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10. What Students Will Learn

With this Docker setup, they will practice:

- Edge computing pipelines
- MQTT communication
- AI at the edge
- IoT dashboards
- Industrial-style system integration